

Dietary nitrate supplementation reduces the O₂ cost of walking and running: a placebo-controlled study.



Dietary supplementation with beetroot juice (BR) has been shown to reduce resting blood pressure and the O₂ cost of submaximal exercise and to increase tolerance to high-intensity cycling. We tested the hypothesis that the physiological effects of BR were consequent to its high NO₃⁻ content per se, and not the presence of other potentially bioactive compounds. We investigated changes in blood pressure, mitochondrial oxidative capacity (Q_{max}), and physiological responses to walking and moderate- and severe-intensity running following dietary supplementation with BR and NO₃⁻-depleted BR [placebo (PL)]. After control (nonsupplemented) tests, nine healthy, physically active male subjects were assigned in a randomized, double-blind, crossover design to receive BR (0.5 l/day, containing ~6.2 mmol of NO₃⁻) and PL (0.5 l/day, containing ~0.003 mmol of NO₃⁻) for 6 days. Subjects completed treadmill exercise tests on days 4 and 5 and knee-extension exercise tests for estimation of Q_{max} (using ³¹P-magnetic resonance spectroscopy) on day 6 of the supplementation periods. Relative to PL, BR elevated plasma NO₂⁻ concentration (183 ± 119 vs. 373 ± 211 nM, P < 0.05) and reduced systolic blood pressure (129 ± 9 vs. 124 ± 10 mmHg, P < 0.01). Q_{max} was not different between PL and BR (0.93 ± 0.05 and 1.05 ± 0.22 mM/s, respectively). The O₂ cost of walking (0.87 ± 0.12 and 0.70 ± 0.10 l/min in PL and BR, respectively, P < 0.01), moderate-intensity running (2.26 ± 0.27 and 2.10 ± 0.28 l/min in PL and BR, respectively, P < 0.01), and severe-intensity running (end-exercise O₂ uptake = 3.77 ± 0.57 and 3.50 ± 0.62 l/min in PL and BR, respectively, P < 0.01) was reduced by BR, and time to exhaustion during severe-intensity running was increased by 15% (7.6 ± 1.5 and 8.7 ± 1.8 min in PL and BR, respectively, P < 0.01). In contrast, relative to control, PL supplementation did not alter plasma NO₂⁻ concentration, blood pressure, or the physiological responses to exercise. These results indicate that the positive effects of 6 days of BR supplementation on the physiological responses to exercise can be ascribed to the high NO₃⁻ content per se.